

SQL ASSIGNMENT

use sakila

//Q1. Write a query to display the first name, last name, and rental date of all customers who rented a film.

//Use the customer and rental tables.

```
select first_name,last_name,rental_date
```

```
from customer C
```

```
INNER join rental R
```

```
on C.customer_id = R.customer_id
```

//Q2. Write a query to display the first name, last name, and address of all customers whose district

//is either Alberta or Texas. Use the customer and address tables.

```
select distinct district from
```

```
select first_name,last_name,address
```

```
from customer C
```

```
inner join address A
```

```
on C.address_id = A.address_id
```

```
and A.district = 'Alberta' or A.district = 'Texas'
```

//Q3. List all films along with their category names, including films that may not belong to any category.

//Then sort it by category names.

//Use the film, film_category and category tables.

```
select distinct category_id from film_category
```

```
select title as film_name ,name as category_name from film f
left join film_category fc
on f.film_id = fc.film_id
left join category c
on fc.category_id = c.category_id
order by 2 asc
```

//Q4. Display the number of rentals per store. Use the rental, inventory, and store tables.

```
select distinct store_id from store
```

```
SELECT s.store_id, COUNT(r.rental_id) AS rental_count
FROM rental r
JOIN inventory i ON r.inventory_id = i.inventory_id
JOIN store s ON i.store_id = s.store_id
GROUP BY s.store_id;
```

// Q5. List the names of categories that have more than 50 films. Use table category and film_category.

```
select c.category_id,name as category_name , COUNT(fc.film_id) as Total_film
from category c
```

```
left join film_category fc
on c.category_id = fc.category_id
group by 1,2
having COUNT(fc.film_id) > 50
```

//Q6. Write a query to display the film titles that have been rented by the customer named 'Mary Smith'.

//Use a subquery to get the customer ID.

```
SELECT f.title
FROM film f
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
WHERE r.customer_id = (
    SELECT customer_id
    FROM customer
    WHERE first_name = 'Mary' AND last_name = 'Smith'
);
```

//Q7. Write a query to display the total rental income generated by each film.

//Use film, inventory, rental and payment tables.

```
SELECT
    f.film_id,
    f.title,
```

```

SUM(p.amount) AS total_rental_income
FROM film f
JOIN inventory i
    ON f.film_id = i.film_id
JOIN rental r
    ON i.inventory_id = r.inventory_id
JOIN payment p
    ON r.rental_id = p.rental_id
GROUP BY f.film_id, f.title
ORDER BY total_rental_income DESC;

```

//Q8. Write a query to display the total number of films each actor has acted in. Use actor, film_Actor tables.

```

select a.actor_id,first_name,last_name,COUNT(fa.film_id) as Total_film
from actor a
left join film_actor fa
on a.actor_id = fa.actor_id
group by 1,2,3
ORDER BY Total_film DESC;

```

//Q9. Write a query to list all rentals with a case statement that displays 'New Customer'

//if the customer_id is greater than 300,

//and 'Regular Customer' otherwise. Use table rental.

```

SELECT

```

```
rental_id,  
customer_id,  
rental_date,  
CASE  
    WHEN customer_id > 300 THEN 'New Customer'  
    ELSE 'Regular Customer'  
END AS customer_type  
FROM rental  
ORDER BY rental_date;
```

//Q10. Write a query to find all films with titles that start with the letter 'A'. Display the film title and

the release year. Use film table and inventory table.

```
SELECT  
    f.title,  
    f.release_year  
FROM film f  
JOIN inventory i  
    ON f.film_id = i.film_id  
WHERE f.title LIKE 'A%';
```

//Q11. Create a new DB and perform below operations using SQL commands.

//(a) Create a table with 2 columns – Student_ID, Marks

```
CREATE TABLE StudentMarks (  
    Student_ID INT PRIMARY KEY,
```

```
Marks INT  
);
```

```
//(b) Insert 2 records in this table.
```

```
INSERT INTO StudentMarks (Student_ID, Marks)  
VALUES  
(1, 85),  
(2, 92);
```

```
//(c) Rename the Marks column with Score.
```

```
ALTER TABLE StudentMarks  
CHANGE Marks Score INT;
```

```
//(d) Then delete the table.
```

```
DROP TABLE StudentMarks;
```

```
//(e) Then delete the Database.
```

```
DROP DATABASE database_name;
```